Section 01

Industrial Security (Master)

IS Security

Section 01 – Firmware Reverse Engineering

Firmware Reverse Engineering

Lecture Notes

Department of Computer Science

Outline – Section 01

1. Why Firmware Matters
2. Sourcing and Triage
3. Deep Analysis
4. Responsible Disclosure

Learning Objectives

By the end of this section you will be able to

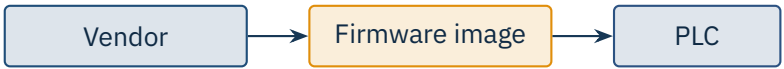
- Source firmware images lawfully and verify their integrity.
- Identify the architecture, bootloader, and root filesystem inside an image.
- Use `binwalk`, `firmware-mod-kit`, and Ghidra to peel a firmware.
- Recognise common backdoors and hard-coded secrets.
- Explain the responsible-disclosure path for ICS findings.

1

Why Firmware Matters

Section 01 – Firmware Reverse Engineering

Why Firmware Reverse Engineering for ICS?



The opaque box

Without firmware analysis the asset owner only sees what the vendor chooses to document. Backdoors, hard-coded credentials, and weak crypto often hide here.

Famous ICS Firmware Findings

- **Project Basecamp (2012)** – 16 PLC families with hard-coded credentials.
- **Schneider M340 (2018)** – Modicon firmware downgrade without signing.
- **Siemens S7-1200 (2019)** – engineering-tool backdoor for bootloader updates.
- **CODESYS Runtime (2023)** – *Code16* suite, 16 CVEs affecting hundreds of products.
- **Boa web server** – abandoned upstream, still in many ICS HMIs (Microsoft, 2022).

Pattern

Long product lifecycles + slow vendor patching + shared third-party code = systemic risk.

2

Sourcing and Triage

Lawfully Acquiring a Firmware Image

- Vendor support portal (per warranty / contract).
- Downloaded from a device you own (read-out tools, JTAG).
- Public update CDN (URLs often discoverable via cached references).
- Manufacturer’s open partner programme (Siemens Industry Online, Rockwell PCDC).

Caution

Compliance Reverse engineering is often legally permitted for security research and interoperability. Check your jurisdiction; in the EU see Directive 2009/24/EC and the CRA’s research safe-harbour.

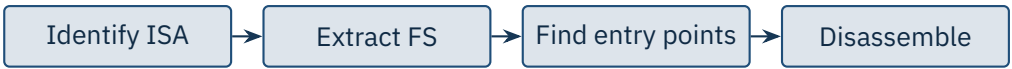
First Pass with `binwalk`


```
$ binwalk -e firmware.bin
DECIMAL     HEX         DESCRIPTION
0           0x0         uImage header, header size: 64 bytes, ...
64          0x40        LZMA compressed data, ...
1572864     0x180000    Squashfs filesystem, little endian, ...
```

Signals

File magic at well-known offsets, recognisable compression algorithms, sane image checksum – and you have probably found bootloader, kernel, and root filesystem.

Triage Workflow



Cheap wins first

Before opening Ghidra, run `strings`, `grep -i pass|key|token`, and look at `/etc/passwd`. Many findings sit on the surface.

3

Deep Analysis

Section 01 – Firmware Reverse Engineering

Architecture Identification

ARM (Cortex-M / A) | MIPS (legacy) | PowerPC | Renesas RX/RH

Detection tools

`file`, `binwalk -A`, `cpu_rec`, and the Ghidra “Auto Analyze” loader. Cross-check against the chip on the actual board if you can.

Hunting Hard-coded Secrets


```
$ strings -n 8 -a firmware.bin | grep -iE 'password|key|secret|admin'
admin
Admin123
SerialPort_default_key=4f6b5e...
$ binwalk -E firmware.bin     # high-entropy regions often hide keys
```

Red flags

Strings like “admin” adjacent to “pass…”, high-entropy 16/32-byte blobs near init code, repeated keys across firmware versions.

Ghidra for ICS Firmware

Workflow tip

For repeat work, save your Ghidra project per vendor family. Common findings (Modbus parser, web admin, OTA verifier) tend to recur across products.

4

Responsible Disclosure

The Disclosure Path

Timelines

Common practice: 90 days vendor + 30 day grace. ICS vendors often need longer (180 days) because of co-ordinated firmware rollouts.

Ethics and Safe-Harbour

- Do not test on production hardware unless it is yours.
- Do not weaponise findings before vendor disclosure is complete.
- Keep proof-of-concept code in a private repo until co-ordinated release.
- Track legal cover (DMCA §1201 in the US, EU CRA research safe-harbour, BSI §202c in Germany).

Caution

Hard line A finding obtained on equipment you do not own (or are not authorised to test) is contaminated – no responsible publication channel will accept it.

Summary – Section 01

★ Key Takeaway: Firmware RE for ICS in one breath

- Source the image lawfully, verify integrity, identify the architecture.
- `binwalk` + `strings` answer 80% of triage questions for free.
- Open Ghidra only after the cheap signals are exhausted.
- Hard-coded credentials and unauthenticated firmware-update paths are still common findings in 2020s ICS devices.
- Coordinated disclosure (vendor + national CERT) is the only ethical publication route.

End of Section 01

Firmware Reverse Engineering

Questions and discussion

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